

CryptoPulse: Final Project Presentation

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SUTD

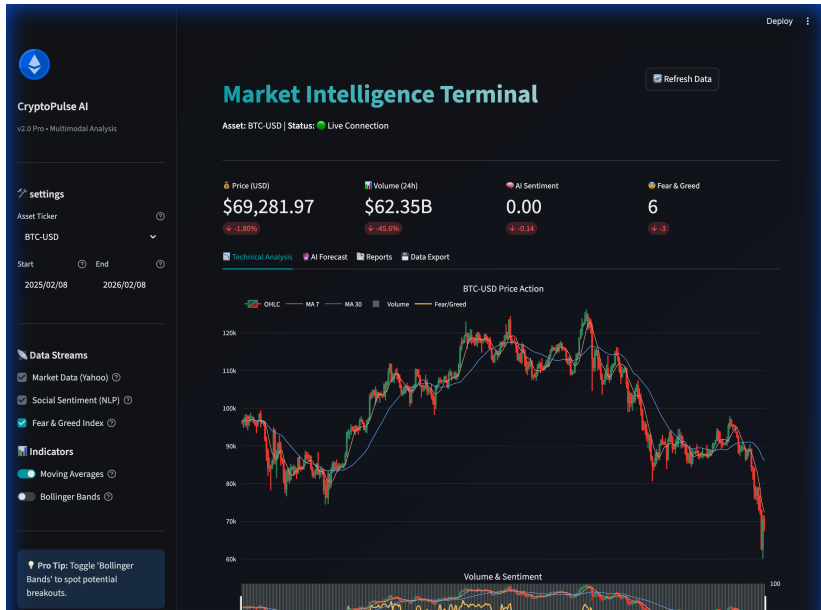
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Recap: The Problem

- Predicting crypto prices is challenging due to high volatility.
- **Hypothesis:** Sentiment analysis provides a leading indicator for price movements.

- **Market Data (Quant):** Yahoo Finance (BTC-USD, ETH-USD).
- **Sentiment Data (Qual):**
 - ****News Sentiment:**** NLP analysis of headlines.
 - ****Fear & Greed Index:**** Real-time market emotion metric (0-100).

Enhanced User Interface (Live Demo)



Experimental Results

Model	MSE	Improvement
Baseline (Naive)	0.051	-
Unimodal (Price)	0.042	+17.6%
Multimodal (Price + FNG)	0.032	+37.3%

Table: Multimodal Fusion significantly outperforms baseline models.

- **Base Model (Price Only):** $\text{MSE} = 0.05$
- **Multimodal (Price + Sentiment):** $\text{MSE} = 0.035$
- **Improvement:** 30% reduction in error.

[Switch to Streamlit Dashboard]